CSC322 – Operating Systems (Lab)

COMSATS University Islamabad Vehari Campus

Course:	***CSC323 — Operating Systems***	***Lab Assignment # 1***
Program / Semester:	***BSSE — 5th Semester (A+B)***	***Due Date: 07-10-2026***
CLO Mapping:	***CLO-5: Operate basic services and functionality of operating systems***	***Total marks: 10***

Instructions

- Perform every task yourself in the Ubuntu terminal of your own virtual machine. Copying another student's work will result in zero marks.
- For each question, submit (1) the exact commands you typed and (2) a screenshot showing the commands and their output in the terminal.
- Where a question asks for an explanation, answer in one or two sentences in your own words.
- Use your own registration number wherever a name needs to be unique (for example, csc322_<RegNo>).
- Submit a single PDF file named LabAssignment1_<RegNo>.

Question 1: Users, Groups and Ownership [5 marks]

Perform the steps below using an account that can use sudo.

(a) Create two users labuser1 and labuser2 with adduser and set a password for each.

(b) Display the UID and GID of both users with id, and display their lines from /etc/passwd.

(c) Switch to labuser1 with su -l labuser1. Copy /bin/ls into the home directory of labuser1 as myls and set its permissions to 0710. Exit back to your own account.

(d) Switch to labuser2 and try to run /home/labuser1/myls. Record the result and explain it.

(e) As the administrator, create a group labgrp, add both users to it, and change the owner and group of myls to labuser1 and labgrp using chown.

(f) Log in as labuser2 again and retry the command from (d). If access is still denied, find the cause, apply the smallest permission change that fixes it, and retry.

(g) Explain in one sentence what the permission value 0710 means for the owner, the group and others.

Deliverable: Commands and screenshots for each step, the result of both attempts in (d) and (f), the cause and fix if needed in (f), and the explanation for (g).

Question 2: Text Processing Utilities and Pipelines [5 marks]

(a) View /etc/passwd page by page with less, scroll forward and backward, and quit.

(b) Use wc on /etc/passwd and state the number of lines, words and bytes. What does the number of lines tell you about the system?

(c) Write one pipeline that creates a single sorted list of the programs in /bin and /usr/bin and shows it page by page.

(d) Write one pipeline that counts the entries of /usr/bin.

(e) Write one pipeline that lists the programs of /bin and /usr/bin whose names contain zip (ignore upper/lower case), sorted alphabetically.

(f) Write one pipeline that shows the first 5 and then (in a separate command) the last 5 entries of the sorted list from (c).

(g) Write one pipeline that shows 10 names from the sorted list of /bin and /usr/bin that do not contain the letter e.

Deliverable: Each command or pipeline with a screenshot of its output, and the answers to (b).